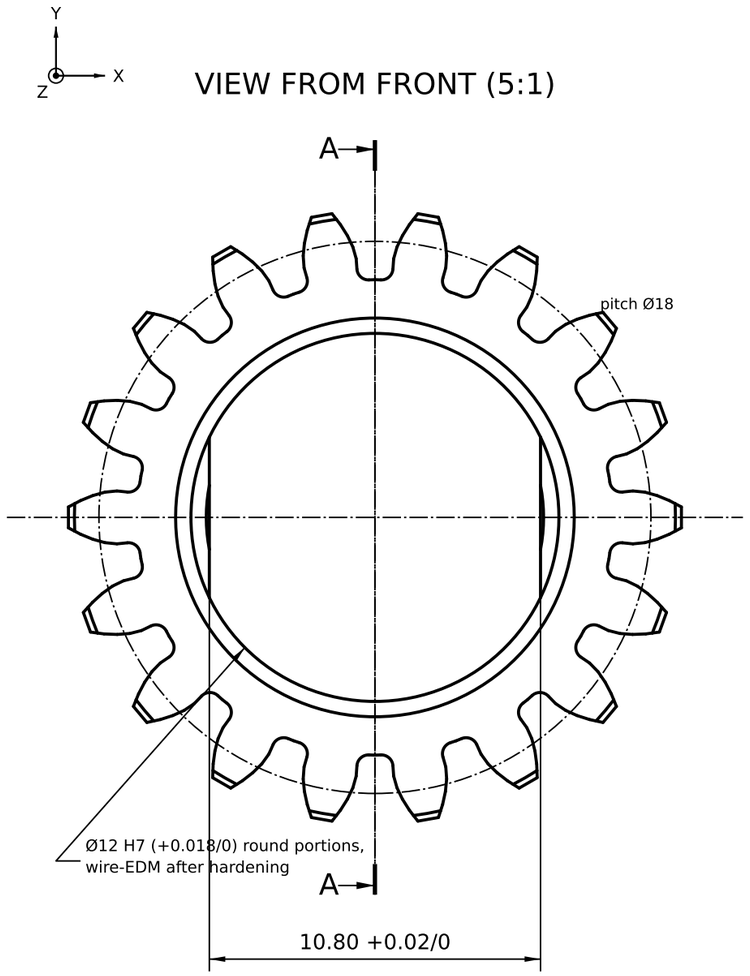
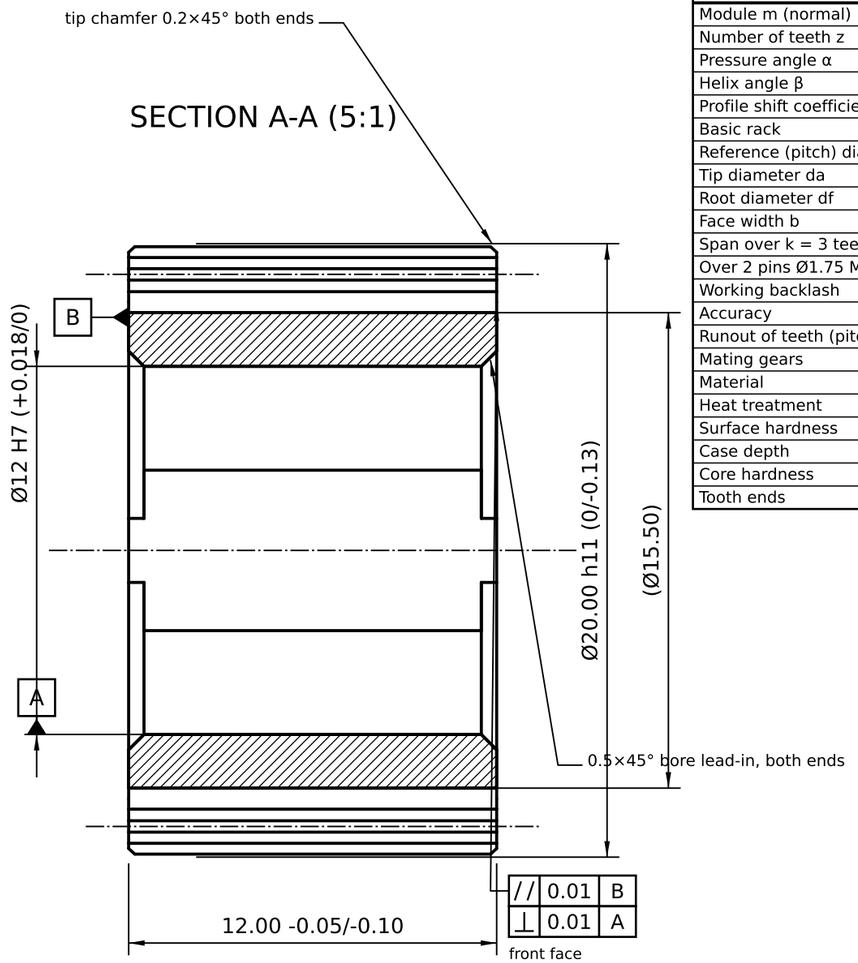
VIEW FROM FRONT (5:1)



SECTION A-A (5:1)



GEAR DATA	
Module m (normal)	1
Number of teeth z	18
Pressure angle α	20°
Helix angle β	0° (spur)
Profile shift coefficient x	0
Basic rack	ISO 53 profile A (ha* 1.0, hf* 1.25, ρf* 0.38)
Reference (pitch) diameter d	18.000
Tip diameter da	20.00 h11 (0/-0.13)
Root diameter df	15.50 (generated)
Face width b	12.00 -0.05/-0.10
Span over k = 3 teeth W3	7.632 nom.; make 7.597 / 7.577 (-0.035/-0.055)
Over 2 pins Ø1.75 Md (alt.)	20.458 nom.; allowance ≈1.5×: -0.053/-0.083 (ref.)
Working backlash	0.05-0.10 at a = 22.500 (tooth thinning above)
Accuracy	ISO 1328-1:2013 grade 6, teeth ground after HT
Runout of teeth (pitch circle) to A	0.01
Mating gears	3 × planet AM-14 (z 27), a = 22.500
Material	18CrNiMo7-6 (1.6587); AISI 9310 / 8620 with cert.
Heat treatment	carburise, quench, temper
Surface hardness	58-62 HRC
Case depth	Eht 0.3-0.5 mm to 550 HV1, after grinding
Core hardness	30-42 HRC
Tooth ends	tip chamfer 0.2×45°

NOTES

1. Dimensions in mm. General tolerances ISO 2768-mK. Fits ISO 286-1/-2 (deviations shown in brackets).
2. STEP file governs undimensioned geometry: cnc/step/sun.step (STEVE 02e31a10 rev 49).
3. Edges ISO 13715 -0.3 (break 0.2-0.4 x 45°) unless a chamfer is shown. Remove all burrs, especially in bores and holes.
4. Surface roughness Ra 3.2 µm unless noted (ISO 21920-1).
5. Teeth are hobbed / shaped and ground by the gear shop to the gear data table; the STEP teeth are a reference shape (ISO 53 profile A with the 0.38 m root fillet), not a machining surface.
6. Bore: double-D, Ø12 H7 round portions, 10.80 +0.02/0 across the flats, wire-EDM after hardening (fits the shaft AM-09 flats 10.80 -0.01/-0.03; slides off forward).
7. Thin rim (1.75 mm under the root): control heat-treat distortion; check the rim is not through-hardened.
8. Inspection: gear report per ISO 1328 (profile, helix, pitch, runout), W3, hardness and case depth on a coupon from the same batch. Supply material certificate.

REV	DESCRIPTION	DATE	BY
B	Tooth thinning for 0.05-0.10 backlash (was 0.03-0.08)	2026-10-17	Claude

PROJECT: **Unknown machine actuator**	DRAWN BY: Claude for macncrash			SOURCE (MODEL GOVERNS): STEVE 02e31a10 rev 49 STEP: cnc/step/sun.step
GENERAL TOLERANCES: ISO 2768-mK (ISO 22081) Edges ISO 13715 -0.3 (0.2-0.4)	PROJECTION: FIRST ANGLE ISO 128 / ISO 5456-2			
FINISH / HEAT TREATMENT: Carburised 58-62 HRC, Eht 0.3-0.5, ground				UNITS: mm
MATERIAL: 18CrNiMo7-6 (1.6587)				SCALE: 5:1
TITLE: **Sun gear m1 18 T**				QTY PER ACTUATOR: **1**
DRAWING / PART NUMBER: **AM-13-sun**	REV: **B**	SHEET: 1/1	SIZE: A3	DATE OF ISSUE: 2026-10-17